INFINITE SERIES

Cauchy's root test:

A positive term series $\sum u_n$

i)Converges, if $\lim\limits_{n\to\infty}(u_n)^{\frac{1}{n}} = \lambda < 1,$

ii)diverges, if $\lim\limits_{n\to\infty}(u_n)^{\frac{1}{n}} = \lambda > 1.$

iii)Test fails, if $\lim\limits_{n\to\infty}(u_n)^{\frac{1}{n}} = \lambda = 1.$

1) $\sum(1+\frac{1}{n})^{-n^2}$

Let $u_n = (1+\frac{1}{n})^{-n^2}$

$$(u_n)^{\frac{1}{n}} = \left(\left(1+\frac{1}{n}\right)^{-n^2}\right)^{1/n}$$

$$= (1+\frac{1}{n})^{-n}$$

$$= \frac{1}{(1+\frac{1}{n})^n}$$

$$\lim_{n\to\infty}(u_n)^{\frac{1}{n}} = \lim_{n\to\infty}\frac{1}{(1+\frac{1}{n})^n}$$

$$= \frac{1}{e} < 1$$

By Cauchy's root test, $\sum u_n$ is convergent.

2) $\sum(1+\frac{1}{\sqrt{n}})^{-n^{3/2}}$

Let $u_n = (1+\frac{1}{\sqrt{n}})^{-n^{3/2}}$

$$(u_n)^{\frac{1}{n}} = \left(\left(1+\frac{1}{\sqrt{n}}\right)^{-n^{3/2}}\right)^{1/n}$$

$$\lim_{n\to\infty}(u_n)^{\frac{1}{n}} = \lim_{n\to\infty}\frac{1}{(1+\frac{1}{\sqrt{n}})^{\sqrt{n}}}$$

$$= \frac{1}{e} < 1$$

By Cauchy's root test, $\sum u_n$ is convergent.

Alternating Series:

A series in which the terms are alternately positive or negative is called an alternating series.

Leibnitz's Rule:

An alternating series $u_1 - u_2 + u_3 - u_4 + \cdots$ converges if

i)Each term is numerically less than its preceding term,

ii) $\lim_{n\to\infty} u_n = 0$.

(Note: if $\lim_{n\to\infty} u_n \neq 0$ then the given series is oscillatory.)

Discuss the convergence of the series

1) $1-\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}-\frac{1}{\sqrt{4}}+\cdots$

Soln The terms of the given series are alternately positive & negative; each term is numerically less than its preceding term &

$$\lim_{n\to\infty} u_n = \lim_{n\to\infty}\frac{1}{\sqrt{n}} = 0$$

Hence by Leibnitz's rule, the given series is convergent.

Examine the character of the series

$$\sum \frac{(-1)^{n-1}\, n}{2n-1}$$

The terms of the given series are alternatively positive & negative ; and each term is numerically less than its preceding term.

But $\lim\limits_{n\to\infty} u_n = \lim\limits_{n\to\infty} \frac{n}{2n-1}$

$$= \lim_{n\to\infty} \frac{1}{2-\frac{1}{n}}$$

$$= \frac{1}{2}, which\ is\ not\ zero.$$

Hence the given series is oscillatory.